# Covariance Estimators and Out-of-Sample Hedging of SOFR Swap Risk*

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## 1 Introduction

In interest rate derivatives, principal component analysis (PCA) hedging neutralizes a portfolio's exposure to the few factors that drive most yield-curve movements. Litterman and Scheinkman (1991) found that three factors, interpreted as level, slope and curvature, explain most of the variation in US Treasury returns, so a hedge against the leading factors needs only as many swaps as factors. The hedge and the risk that remains after it are both computed from an estimated covariance matrix of daily rate changes. This study examines whether the choice of covariance estimator affects the hedging effectiveness of a Secured Overnight Financing Rate (SOFR) swap portfolio and the accuracy of its out-of-sample risk forecasts, using daily rates from 14 swap tenors spanning 2024 to 2025. The portfolio is hedged with one or two swaps, either by neutralizing the leading principal components or by minimizing variance.

The sample covariance matrix weights a year of changes equally. Ledoit and Wolf (2004) shrink it toward a scaled identity matrix with the weight that is asymptotically optimal under a quadratic loss, and Chen et al. (2010) derive the oracle approximating shrinkage (OAS) weight for Gaussian samples. An exponentially weighted moving average (EWMA) instead discounts older changes, and the RiskMetrics methodology uses a decay factor of 0.94 for daily data (J.P. Morgan/Reuters, 1996). Five estimators are compared, being the sample, the two shrinkage estimators and EWMA at two decay factors.

Our analysis indicates that no estimator consistently outperforms the equal-weighted sample in terms of hedging effectiveness. We found that using shrinkage leaves PCA hedges unchanged and actually worsens minimum-variance hedges with two swaps, in up to 38 of the 79 pairs. In contrast, the EWMA method enhances 36 or 37 out of the 79 PCA hedges with two swaps while negatively affecting 6 to 10 of them. The choice of estimator significantly influences the risk forecast. As volatility decreases in 2025, equal weighting tends to overstate the unhedged risk. EWMA with a decay factor of 0.94 shows no significant bias. However, shrinkage tends to overstate the residual risk of hedged portfolios by a factor of 2.5 to 4.

The remainder of the report is organized as follows. Section 2 describes the data. Section 3 states the hedges, the estimators and the evaluation. Section 4 reports the results, and Section 5 concludes.[1]

## 2 Data

Daily SOFR swap rates at the 14 tenors of 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 15, 20, 25 and 30 years are collected from 2 January 2024 to 31 December 2025. Two holidays with partial quotes, 19 February and 4 July 2024, are removed, and daily changes are taken in basis points (bp). Table 1 reports the first principal component (PC1) of the changes by half-year. PC1 explains 93% to 95% of the variance in every half-year, and its daily standard deviation falls from about 20 bp to 13.7 bp in the second half of 2025.

*This project builds on homework for Fordham University's QFGB 8934 course.

[1]Code and replication data are available at *GitHub*.

| Half-year | PC1 standard deviation (bp) | PC1 share of variance (%) |
|---|---|---|
| 2024 H1 | 21.4 | 95.3 |
| 2024 H2 | 19.5 | 93.0 |
| 2025 H1 | 20.5 | 93.7 |
| 2025 H2 | 13.7 | 93.2 |

Table 1: PC1 of daily SOFR swap rate changes by half-year.

# 3 Methodology

## 3.1 Hedges

Write $\Delta \boldsymbol{y}_t \in \mathbb{R}^p$ ($p = 14$) for the vector of daily rate changes on day $t$, and $\boldsymbol{d}$ for the dollar value of 1 bp (DV01) of the portfolio by tenor, in dollars per bp. The portfolio is

$$\boldsymbol{d} = (-5.0, -2.0, -1.0, 3.5, -0.1, 1.0, 2.3, -0.8, -0.6, 2.5, 3.0, 4.0, 1.5, 5.0)^\top \times 10^3, \tag{1}$$

ordered from the 2-year to the 30-year tenor, with a net DV01 of \$13,300. It is hedged with $k \in \{1, 2\}$ swaps at the tenors selected by the matrix $B \in \mathbb{R}^{p \times k}$, whose columns are unit vectors, and $\boldsymbol{h}$ holds their DV01s. The hedged DV01 and its profit and loss (P&L) on the next day are

$$\boldsymbol{q} = \boldsymbol{d} + B\boldsymbol{h}, \qquad \pi_{t+1} = -\boldsymbol{q}^\top \Delta \boldsymbol{y}_{t+1}. \tag{2}$$

Let $\hat{\Sigma}$ be a covariance estimate with eigenvectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$ ordered by decreasing eigenvalue, and $V_k = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)$. The PCA hedge sets the exposure of $\boldsymbol{q}$ to the $k$ leading components to zero, and the minimum-variance hedge minimizes $\boldsymbol{q}^\top \hat{\Sigma} \boldsymbol{q}$,

$$\boldsymbol{h}_{\text{PCA}} = -(V_k^\top B)^{-1} V_k^\top \boldsymbol{d}, \qquad \boldsymbol{h}_{\text{MV}} = -(B^\top \hat{\Sigma} B)^{-1} B^\top \hat{\Sigma} \boldsymbol{d}. \tag{3}$$

The hedge sets are the 14 single tenors and the 79 of 91 pairs whose condition number $\kappa(V_2^\top B)$ is below 10 in the first estimation window, since an ill-conditioned pair turns small errors in $V_2$ into large hedges.

## 3.2 Covariance Estimators

Five estimators are compared in this study. The equal-weighted estimator is the sample covariance matrix $S$ of the $W = 252$ daily changes before day $t$. The estimator of Ledoit and Wolf (2004), hereafter LW, and the OAS estimator of Chen et al. (2010) shrink it toward a scaled identity matrix,

$$\hat{\Sigma} = (1 - \delta) S + \delta \tau I, \qquad \tau = \operatorname{tr}(S)/p, \tag{4}$$

with the intensities $\delta$ given in Appendix A. The EWMA estimator applies the covariance recursion of J.P. Morgan/Reuters (1996) to all changes before day $t$, with the weights normalized to sum to one,

$$\hat{\Sigma} = \sum_{j \geq 1} w_j \, \Delta \boldsymbol{y}_{t-j} \Delta \boldsymbol{y}_{t-j}^\top, \qquad w_j \propto \lambda^{j-1}. \tag{5}$$

The decay factor $\lambda$ is 0.94, the RiskMetrics value for daily data, or 0.98906, which gives a half-life of 63 trading days.

## 3.3 Evaluation

Every test day $t$ from 3 January to 31 December 2025 is hedged with the estimates from the changes before it, and the P&L $\pi_t$ of (2) is recorded. Hedge effectiveness is measured by the residual risk

$$\text{RR} = \frac{\sum_t \pi_{t,\text{hedged}}^2}{\sum_t \pi_{t,\text{unhedged}}^2}, \tag{6}$$

reported as the median over each hedge set. For each hedge, the squared P&L under each estimator is compared with that under the equal-weighted estimator by the test of Diebold and Mariano (1995), and the hedges significantly better or worse at the 5% level are counted. The risk forecast is evaluated by

$$z_t = \frac{\pi_t}{\sqrt{\boldsymbol{q}^\top \hat{\Sigma} \boldsymbol{q}}}, \qquad \mathrm{H}_0 \colon \ \mathbb{E}\left[z_t^2\right] = 1, \tag{7}$$

and the mean $\overline{z^2}$ of $z_t^2$ over the test days falls below one when the estimator overstates the risk. All tests use the $t$ statistic $t_{\mathrm{NW}}$ with the variance estimator of Newey and West (1987), stated in Appendix A.

# 4 Results

## 4.1 Hedge Ratios

Table 2 shows that a single swap can reduce the squared P&L by 75% to 77%, while two swaps can eliminate 97% to 98%, regardless of the estimator used. With just one swap, both rules result in similar residual risk because PC1 accounts for most of the variance. No estimator outperforms the equal-weighted sample. Additionally, the shrinkage method yields the same PCA hedges as the equal-weighted estimator, as demonstrated in Proposition 1 below.

| Rule | Swaps | Equal | LW | OAS | EWMA 0.94 | EWMA 0.98906 |
|---|---|---|---|---|---|---|
| PCA | 1 | 22.9 | 22.9 | 22.9 | 24.8 | 23.2 |
| | 2 | 2.56 | 2.56 | 2.56 | 2.32 | 2.34 |
| Minimum variance | 1 | 23.0 | 23.0 | 23.0 | 24.7 | 23.3 |
| | 2 | 2.16 | 2.93 | 2.69 | 2.18 | 2.09 |

Table 2: Out-of-sample residual risk (%), median over 14 single tenors or 79 pairs.

Table 3 tests the differences hedge by hedge. Shrinkage makes the minimum-variance hedge with two swaps worse on 38 pairs under LW and 30 under OAS, and better on none. Its effect follows from (4),

$$B^\top \hat{\Sigma} B = (1-\delta) B^\top S B + \delta\tau I, \qquad B^\top \hat{\Sigma} \boldsymbol{d} = (1-\delta) B^\top S \boldsymbol{d} + \delta\tau \boldsymbol{d}_B, \tag{8}$$

where $\boldsymbol{d}_B = B^\top \boldsymbol{d}$ is the DV01 at the hedge tenors. The identity target alone would give $\boldsymbol{h}_{\mathrm{MV}} = -\boldsymbol{d}_B$, a hedge that offsets only the DV01 at the hedge tenors and ignores the correlation across the curve, and shrinkage pulls the hedge toward it. For two swaps whose rates move almost together, the smaller eigenvalue of $B^\top S B$ is small, so adding $\delta\tau$ to it changes the hedge materially. EWMA changes almost no minimum-variance hedge with two swaps, and with a decay factor of 0.94 it is worse on 4 of the 14 single tenors under both rules. For PCA hedges with two swaps, it is better on 36 or 37 pairs and worse on 6 to 10.

## 4.2 Risk Forecasts

Table 4 illustrates the impact of the estimator. For the unhedged portfolio, the equal-weighted, LW, and OAS forecasts tend to overestimate the risk, with $\overline{z^2}$ around 0.73. This occurs because the rolling window continues to reflect the volatility from the previous year, while the volatility of PC1 decreases in 2025. The EWMA method with a decay factor of 0.94 tracks this decline accurately and exhibits no significant bias. In contrast, a decay factor of 0.98906 has minimal effect on reducing this bias.

| | | | PCA | | Minimum variance | |
|---|---|---|---|---|---|---|
| Swaps | Estimator | Hedges | Better | Worse | Better | Worse |
| 1 | LW | 14 | – | – | 0 | 0 |
| | OAS | 14 | – | – | 0 | 0 |
| | EWMA 0.94 | 14 | 0 | 4 | 0 | 4 |
| | EWMA 0.98906 | 14 | 0 | 0 | 0 | 0 |
| 2 | LW | 79 | – | – | 0 | 38 |
| | OAS | 79 | – | – | 0 | 30 |
| | EWMA 0.94 | 79 | 37 | 10 | 0 | 0 |
| | EWMA 0.98906 | 79 | 36 | 6 | 1 | 0 |

Table 3: Hedges with significantly lower (better) or higher (worse) squared P&L than under the equal-weighted estimator, by the test of Diebold and Mariano (1995) at the 5% level. Under LW and OAS, PCA hedges equal those of the equal-weighted estimator and are not tested.

For hedged portfolios, the equal-weighted forecast shows no significant bias. However, with EWMA at a decay factor of 0.94, the realized variance exceeds the forecast by approximately 24% to 25%, although this discrepancy is not statistically significant. Lastly, shrinkage in forecasting tends to overstate the risk for hedged portfolios by a factor of about 2.5 to 4.

| | Unhedged | | PCA, 2 swaps | | Minimum variance, 2 swaps | |
|---|---|---|---|---|---|---|
| Estimator | $\overline{z^2}$ | $t_{\mathrm{NW}}$ | $\overline{z^2}$ | $t_{\mathrm{NW}}$ | $\overline{z^2}$ | $t_{\mathrm{NW}}$ |
| Equal | 0.732 | −3.24 | 1.03 | 0.22 | 0.964 | −0.30 |
| LW | 0.737 | −3.17 | 0.237 | −24.7 | 0.388 | −9.05 |
| OAS | 0.735 | −3.19 | 0.296 | −18.4 | 0.408 | −9.23 |
| EWMA 0.94 | 0.961 | −0.44 | 1.24 | 1.70 | 1.25 | 1.73 |
| EWMA 0.98906 | 0.770 | −2.83 | 0.984 | −0.13 | 0.948 | −0.44 |

Table 4: Calibration of the risk forecast, with $t_{\mathrm{NW}}$ for $\mathrm{H}_0\colon\ \mathbb{E}[z_t^2] = 1$. For hedged portfolios, $z_t^2$ is averaged over the 79 pairs on each day.

Proposition 1 outlines both effects of shrinkage. The proof can be found in Appendix B.

**Proposition 1.** *Let $\hat{\Sigma} = (1-\delta)S + \delta\tau I$ with $0 \le \delta < 1$ and $\tau = \mathrm{tr}(S)/p$, and let $S$ have distinct eigenvalues. Then*

*(i) $\hat{\Sigma}$ and $S$ have the same eigenvectors in the same order, so the PCA hedge is the same under both, and*

*(ii) for every $\boldsymbol{q}$ with $\boldsymbol{q}^\top S\boldsymbol{q} > 0$,*

$$\textit{Inflation} = \frac{\boldsymbol{q}^\top \hat{\Sigma}\boldsymbol{q}}{\boldsymbol{q}^\top S\boldsymbol{q}} = (1-\delta) + \frac{\delta\tau}{\rho(\boldsymbol{q})}, \qquad \rho(\boldsymbol{q}) = \frac{\boldsymbol{q}^\top S\boldsymbol{q}}{\boldsymbol{q}^\top\boldsymbol{q}}. \tag{9}$$

Part (i) holds because shrinkage toward a scaled identity moves every eigenvalue toward their grand mean (Ledoit and Wolf, 2004) and leaves the eigenvectors unchanged. Part (ii) shows that the inflation of the forecast variance depends on the Rayleigh quotient $\rho(\boldsymbol{q})$, the variance of the portfolio per unit of squared DV01. An unhedged portfolio is dominated by the level factor and has a large $\rho(\boldsymbol{q})$, so the inflation is negligible. A hedged portfolio has its exposure to the leading components removed and a small $\rho(\boldsymbol{q})$, so the term $\delta\tau/\rho(\boldsymbol{q})$ inflates its forecast substantially, as confirmed by Table 5.

| | | Inflation | |
|---|---|---|---|
| Portfolio | $\rho(\boldsymbol{q})$ (bp$^2$) | LW | OAS |
| Unhedged | 52.7 | 0.994 | 0.996 |
| PCA, 2 swaps | 0.108 | 4.47 | 3.43 |

Table 5: Median Rayleigh quotient and median inflation over the test days, and for PCA hedges also over the 79 pairs.

## 5 Conclusion

This study evaluates whether covariance estimator selection impacts the out-of-sample hedging and risk-forecasting performance of a 14-tenor SOFR swap portfolio across 2025. We find that advanced estimators offer no consistent hedging improvement over the standard equal-weighted sample. Shrinkage estimators leave PCA hedges unchanged while degrading two-swap minimum-variance hedges by pulling allocations toward offsetting DV01 strictly at the hedge tenors. Conversely, EWMA improves 36 to 37 of the 79 two-swap PCA hedges and worsens only 6 to 10.

While hedging performance is largely robust to estimator choice, risk forecasting is highly sensitive. Equal weighting overstates unhedged risk during a falling volatility regime, whereas EWMA ($\lambda = 0.94$) shows no significant bias. In contrast, shrinkage estimators overstate hedged residual risk by a factor of 2.5 to 4 due to their distortion of small eigenvalues, to which a hedged portfolio is heavily exposed.

It should be noted that these findings are drawn from a single portfolio over a one-year period, so the ranking of the estimators for the risk forecast may shift in high-volatility regimes. Expanding the historical dataset across multiple economic cycles remains an important avenue for future research.

## References


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# A Definitions

Write $\boldsymbol{x}_s$ for the demeaned changes in the estimation window, $s = 1, \dots, W$, and $\|A\|_F^2 = \operatorname{tr}(AA^\top)$. The LW intensity is the feasible weight of Ledoit and Wolf (2004) on the scaled identity, and the OAS intensity is that of Chen et al. (2010),

$$\delta_{\mathrm{LW}} = \frac{\min\left(\bar{b}^2, \|S - \tau I\|_F^2\right)}{\|S - \tau I\|_F^2}, \qquad \bar{b}^2 = \frac{1}{W^2}\sum_{s=1}^{W} \left\|\boldsymbol{x}_s\boldsymbol{x}_s^\top - S\right\|_F^2, \tag{10}$$

$$\delta_{\mathrm{OAS}} = \min\left(1, \frac{\operatorname{tr}(S^2) + \operatorname{tr}^2(S)}{(W+1)\big(\operatorname{tr}(S^2) - \operatorname{tr}^2(S)/p\big)}\right). \tag{11}$$

The condition number is $\kappa(A) = \sigma_{\max}(A)/\sigma_{\min}(A)$, the ratio of the largest to the smallest singular value. For a daily series $g_t$, $t = 1, \dots, T$, with mean $\bar{g}$, the test statistic uses the variance estimator of Newey and West (1987),

$$t_{\mathrm{NW}} = \frac{\bar{g}}{\sqrt{\hat{\Omega}/T}}, \qquad \hat{\Omega} = \hat{\gamma}_0 + 2\sum_{l=1}^{L}\Big(1 - \frac{l}{L+1}\Big)\hat{\gamma}_l, \tag{12}$$

where $\hat{\gamma}_l$ is the sample autocovariance of $g_t$ at lag $l$. The lag is set to $L = \lfloor 4(T/100)^{2/9} \rfloor$, which is 4 for $T = 251$. The test of Diebold and Mariano (1995) applies it to $g_t = \pi_{t,e}^2 - \pi_{t,\mathrm{Equal}}^2$ for estimator $e$, and the calibration test to $g_t = z_t^2 - 1$.

# B Proof of Proposition 1

*Proof.* Let $S\boldsymbol{v}_i = \ell_i \boldsymbol{v}_i$ with $\ell_1 > \cdots > \ell_p$. Then

$$\hat{\Sigma}\boldsymbol{v}_i = (1-\delta)\ell_i\boldsymbol{v}_i + \delta\tau\boldsymbol{v}_i = \big((1-\delta)\ell_i + \delta\tau\big)\boldsymbol{v}_i, \tag{13}$$

so $\boldsymbol{v}_i$ is an eigenvector of $\hat{\Sigma}$. Since $\delta < 1$, the map $\ell \mapsto (1-\delta)\ell + \delta\tau$ is strictly increasing and preserves eigenvalue order. Hence $V_k$ is the same under $\hat{\Sigma}$ and $S$ up to the signs of its columns. Since $(DV_k^\top B)^{-1}DV_k^\top \boldsymbol{d} = (V_k^\top B)^{-1}V_k^\top \boldsymbol{d}$ for any $D = \operatorname{diag}(\pm 1)$, the signs do not change $\boldsymbol{h}_{\mathrm{PCA}}$ in (3), which depends on the estimate only through $V_k$. For part (ii),

$$\boldsymbol{q}^\top\hat{\Sigma}\boldsymbol{q} = (1-\delta)\,\boldsymbol{q}^\top S\boldsymbol{q} + \delta\tau\,\boldsymbol{q}^\top\boldsymbol{q}, \tag{14}$$

and dividing by $\boldsymbol{q}^\top S\boldsymbol{q}$ gives (9). □